AN ALGEBRAIC EXTENSION GENERATED BY ELEMENTS WHOSE MINIMAL POLYNOMIALS SPLIT COMPLETELY IS NORMAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The displayed set equality (source lines 2150-2153) is the heart of the proof; the hypothesis 'splits completely in E' is what makes every alpha_{i,j} an element of E so that the embeddings apply.

Editorial note. The label lemma-normally-generated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-characterize-normal (normality via the equal-image criterion: E/F is normal iff any two F-embeddings of E into an algebraic closure have the same image)

Source reference. Original reference 09HQ: sources/stacks-fields/pinned-original-source.tex, source lines 2096–2096. Prerequisite pointer invoked at 2159; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be an algebraic extension of fields. If E is generated by $\alpha_i \in E$, $i \in I$ over F and if for each i the minimal polynomial of α_i over F splits completely in E, then E/F is normal.*

Proof. Let P_i be the minimal polynomial of α_i over F. Let $\alpha_i = \alpha_{i,1}, \alpha_{i,2}, \ldots, \alpha_{i,d_i}$ be the roots of P_i over E. Given two embeddings $\sigma, \sigma' : E \to \overline{F}$ over F we see that

$$\{\sigma(\alpha_{i,1}), \ldots, \sigma(\alpha_{i,d_i})\} = \{\sigma'(\alpha_{i,1}), \ldots, \sigma'(\alpha_{i,d_i})\}$$

because both sides are equal to the set of roots of P_i in $\overline{F}$. The elements $\alpha_{i,j}$ generate E over F and we find that $\sigma(E) = \sigma'(E)$. Hence E/F is normal by Lemma 09HQ. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.